Treatments for Measles

First Line

- Supportive care
- Treatment of secondary infections
- Vitamin A supplementation
- Immune globulin, IM[a]
- Measles vaccine

Second Line

- Ribavirin[a]

[a] In select cases, please see text.

Vitamin A supplementation is recommended for all children with measles who live in communities where vitamin A deficiency is known or where measles fatality rate is 1 percent. In the United States, supplementation is considered for children aged 6 months to 2 years who are hospitalized with measles and its complications, or for children older than 6 months who are at risk for vitamin A deficiency (e.g., immunodeficiency, severe malnutrition). More recent studies of vitamin A have led to some controversy regarding its use.

All individuals at risk—children less than 1 year of age, pregnant women, immunocompromised individuals, and those exposed to measles virus—should receive immunoglobulin prophylaxis within 6 days of exposure. If administered within 72 hours of exposure, infection can be prevented. Healthy individuals should receive 0.25 mL/kg of intramuscular immunoglobulin, while immunocompromised patients require 0.5 mL/kg. Exposed patients (excluding pregnant women and those with impaired immune systems) should also receive the measles vaccine 5 months later to confer lasting protection.

Ribavirin has been studied in patients with subacute sclerosing panencephalitis, but no benefit was noted in clinical trials.

Prevention (Immunizations)

The incidence of measles has decreased worldwide as a direct result of immunization. A single dose of the live attenuated measles vaccine produces detectable antibody levels in 95% of individuals, conferring lifelong immunity.

Common side effects of the measles vaccine include fever and transient morbilliform rash, which resolve without treatment. Less common adverse reactions include thrombocytopenia and transient neurologic events.

Measles vaccine is contraindicated in individuals with moderate to severe illness, allergy to eggs or neomycin, pregnant women, and those with impaired immune systems (e.g., cancer, non-HIV immunodeficiency, immunosuppressive therapy). Patients may receive the vaccine 3 months after discontinuation of chemotherapy or immunosuppressive medications.

There have been many unconfirmed and unsubstantiated reports suggesting a link between the measles-mumps-rubella (MMR) vaccine and autism or inflammatory bowel disease. Current scientific evidence does not support a causal relationship between MMR vaccine and autism or inflammatory bowel disease.

Rubella

Epidemiology

Rubella virus has a worldwide distribution, with outbreaks most frequent in late winter and early spring. School-aged children, adolescents, and young adults are most commonly affected. Epidemics occasionally occur in developing countries, particularly where vaccine access is limited.

Since the introduction of the rubella vaccine in the United States in 1969, the incidence of rubella and congenital rubella syndrome has declined dramatically, with no widespread epidemics reported. Occasional outbreaks are largely due to failure to vaccinate susceptible individuals.

Etiology and Pathogenesis

Rubella is an enveloped RNA virus in the *Togaviridae* family. It is transmitted through direct or droplet contact with nasopharyngeal secretions.

Rubella: At a Glance

- Also known as German measles or 3-day measles
- Epidemic disease with worldwide distribution
- Short prodrome; rash lasts 2 to 3 days
- Characteristic enlargement of cervical, suboccipital, and postauricular lymph nodes
- High risk of fetal malformations with congenital infection (e.g., microcephaly, congenital heart disease, deafness), especially during the first trimester